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Microbial Oxidative Stress in Food Fermentations: A Comprehensive Review

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ABSTRACT

This review provides a comprehensive analysis of microbial oxidative stress in food fermentations, establishing it as a central factor that governs microbial physiology, process efficiency, and final product quality. We systematically dissect the dual nature of oxidative stress by first detailing its endogenous sources and exogenous drivers. The review then explores its dichotomous impact, contrasting the role of low-level reactive oxygen species (ROS) as essential signaling molecules that trigger adaptive defenses, with the detrimental effects of excessive ROS accumulation that causes widespread macromolecular damage and inhibits fermentation. We summarize the sophisticated sensing and regulatory networks in bacteria and fungi that orchestrate multilayered defenses, including enzymatic scavenging, redox buffering, and damage repair systems. A key insight is that oxidative stress can be viewed as a process-dependent redox variable that may be monitored and modulated to improve fermentation robustness and product quality. In conclusion, oxidative stress represents a redox dimension with strong potential for process management, but its practical use requires process-specific validation linking redox indicators with fermentation performance and product quality. Integrating validated strain improvement strategies with process-specific control may help mitigate cellular damage while preserving beneficial stress responses, thereby supporting more robust, efficient, and high-quality food fermentations.

1 | Introduction

Microbial fermentation is central to the production of fermented foods and beverages, where microbial activity determines process efficiency, sensory quality, nutritional attributes, and product stability. Advances in microbial physiology show that oxidative stress is a key factor regulating microbial function and fermentation outcomes (Storz and Imlay 1999; Q. Li et al. 2009; Feng and Wang 2020). This understanding has evolved from viewing oxidative stress merely as

cellular damage to recognizing its complex regulatory roles that extend far beyond simple toxicity to encompass sophisticated signaling networks governing microbial robustness, fermentation performance, and product quality in food systems.

Oxidative stress occurs when reactive oxygen species (ROS) like superoxide anion, hydrogen peroxide, and hydroxyl radicals exceed the cell's ability to neutralize them (Imlay 2013; Sies 2017). In fermentation systems, ROS originates from both

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endogenous metabolic processes and exogenous factors such as oxygen supply, agitation intensity, temperature fluctuations, and inhibitory compounds present in raw materials (Enfors et al. 2001). While elevated ROS levels can damage critical cellular components including lipids, proteins, and DNA, controlled low concentrations function as essential signaling molecules that regulate gene expression, coordinate stress responses, and control physiological processes including morphogenesis and secondary metabolite production (Enfors et al. 2001; Sies 2017; Heller and Tudzynski 2011).

This dual nature of oxidative stress represents both a challenge and an opportunity in fermentation technology. Oxidative stress acts simultaneously as a disruptor and regulator in microbial fermentation systems (D'Autr aux and Toledano 2007; Sies 2017). When uncontrolled, it can significantly reduce process efficiency, product yields, and final product quality, creating substantial challenges for industrial-scale operations. However, understanding the mechanisms underlying oxidative stress and developing strategies for its effective management enables significant improvements in microbial strain resilience, process control precision, and targeted metabolic engineering approaches. Recent technological advances in multi-omics analysis, real-time monitoring systems, and synthetic biology tools now provide researchers with unprecedented capabilities to examine these complex interactions at molecular, cellular, and process levels (Mandenius and Gustavsson 2015; Nielsen and Keasling 2016; Chen and Nielsen 2019).

Given the crucial and multifaceted role that oxidative stress plays in fermentation systems, this comprehensive review aims to clarify the diverse sources of oxidative stress, examine its contrasting effects on microbial physiology and process outcomes, and provide detailed analysis of detection methods, cellular response strategies, and practical applications across various fermentation systems. The review systematically covers both endogenous and exogenous origins of oxidative stress, discusses the dichotomous impacts on microbial metabolism and fermentation performance, and summarizes recent findings on how microorganisms detect and respond to oxidative signals through sophisticated regulatory networks. Additionally, we explore key mitigation strategies and practical engineering solutions for controlling oxidative stress, concluding with current challenges and future research directions that will shape the field's development.

A central message of this review is that oxidative stress in food fermentation should not be viewed only as a harmful factor to

be eliminated. Rather, it represents a process-dependent redox variable that can be monitored, modulated, and, under controlled conditions, potentially exploited to improve microbial robustness, fermentation stability, and product quality. However, such management requires process-specific validation because redox indicators and optimal redox windows vary among microorganisms, substrates, fermentation stages, and target quality attributes.

This review focuses on food fermentation systems rather than general industrial biomanufacturing. General microbial physiology is discussed only when it helps explain redox-dependent microbial performance, community dynamics, process stability, or product quality in fermented foods.

2 | Sources of Oxidative Stress in Fermentation Systems

Oxidative stress in fermentation systems arises from multiple interconnected sources that can be broadly categorized into endogenous and exogenous factors (Figure 1). Understanding these sources is crucial for developing effective management strategies that optimize fermentation performance while minimizing cellular damage.

ROS mainly arise during cellular respiration, making it a major source of oxidative stress for aerobic and facultative anaerobic microorganisms (Imlay 2013). In oxidative phosphorylation, electrons from nutrients travel through the electron transport chain—located in the inner mitochondrial membrane of eukaryotes and the cytoplasmic membrane of prokaryotes—to oxygen, generating a proton gradient that powers ATP synthesis (Turrens 2003). Electron transfer in the ETC is imperfect, with roughly 1%–2% of electrons leaking at Complexes I and III, which reduces oxygen to superoxide anion radicals ($O_2^{\cdot-}$), the main ROS. These can further convert into other ROS such as hydrogen peroxide, either spontaneously or via superoxide dismutase (SOD) (Fridovich 1995). This issue is particularly relevant in industrial fermentation, where maximizing microbial growth and productivity is essential. In food fermentations that require aerobic or microaerobic metabolism, such as acetic acid fermentation, mold-based koji fermentation, and surface-ripened cheese production, increased respiratory activity may enhance electron leakage and expose microorganisms to persistent oxidative pressure, thereby requiring robust antioxidant defenses (Meyer et al. 2011).

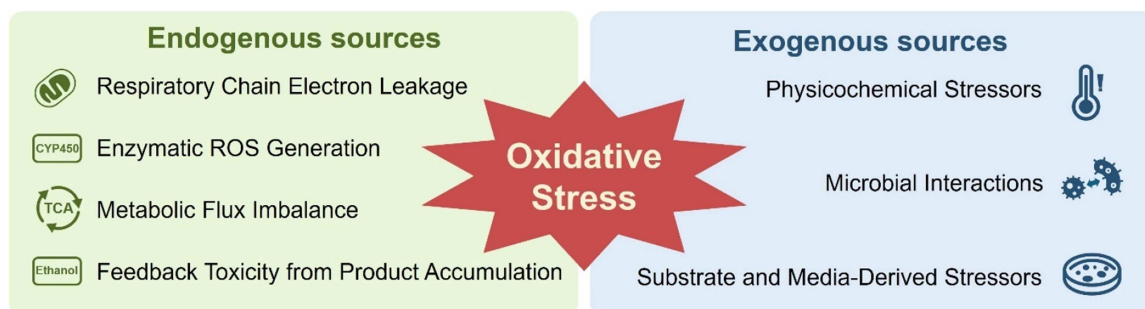


FIGURE 1 | Sources of oxidative stress in fermentation systems.

Redox-active enzymes, alongside the electron transport chain, contribute significantly to ROS production in microorganisms (Scialo et al. 2013; D'Autréaux and Toledano 2007). These enzymes aid metabolic adaptation, growth, and secondary metabolism in both prokaryotes and eukaryotes (Aguirre et al. 2005; Tamano 2014). NADPH oxidases (NOX) and cytochrome P450 monooxygenases are significant families, with NOX generating superoxide anions by transferring electrons from NADPH to oxygen (Tudzynski et al. 2012; Sumimoto 2008). In fungi like *Aspergillus nidulans* and *Neurospora crassa*, NOX influences development, differentiation, and secondary metabolite synthesis (Cano-Domínguez et al. 2008; Zhao et al. 2020), acting as signaling molecules and aiding stress defense (Heller and Tudzynski 2011; Burtenshaw et al. 2017). Cytochrome P450 monooxygenases introduce oxygen into various substrates, creating ROS such as superoxide and hydrogen peroxide due to partial reduction during catalytic cycles (Munro et al. 2007; Davydov et al. 2017). Other oxidases, including xanthine oxidase and glucose oxidase, further contribute to ROS levels, especially under metabolic stress (Dijkman et al. 2013).

Endogenous ROS also arises from metabolic imbalances, especially under high-flux industrial fermentation conditions (Zielinski 2015). When glycolysis outpaces the downstream oxidative pathways like the TCA cycle and electron transport chain (Sanchez and Demain 2002; Folch et al. 2021), metabolic bottlenecks form and disrupt cellular redox balance, increasing the NADH/NAD⁺ ratio and promoting ROS formation (Herst and Berridge 2006). The Crabtree effect in yeasts such as *Saccharomyces cerevisiae* exemplifies this phenomenon: high glucose levels shift carbon flux toward fermentation and ethanol production even with available oxygen, due to saturated mitochondrial respiratory capacity (Perez-Samper et al. 2018; Zhang et al. 2022). This process increases cytoplasmic NADH, facilitating ROS generation (Sussman et al. 1980). Similarly, spike feeding during fed-batch fermentations can trigger overflow metabolism and ROS bursts (Valentinotti et al. 2003). Critical enzymatic hubs like pyruvate dehydrogenase and α -ketoglutarate dehydrogenase are major ROS sources under catalytic overload (Martínez-Reyes and Chandel 2020). Their E3 subunits contain FAD cofactors that leak electrons to oxygen if there is substrate or coenzyme imbalance (Tretter and Adam-Vizi 2005).

Product accumulation during fermentation often results in feedback toxicity that amplifies oxidative stress and limits further production (Sanchez and Demain 2008). In ethanol fermentation, accumulating ethanol compromises cell membranes and impairs mitochondrial function, disrupting electron transport and increasing ROS such as superoxide (Huffer et al. 2011). This ROS production leads to mitochondrial damage and apoptotic death, directly linking product buildup to oxidative stress (Farrugia and Balzan 2012). Organic acid fermentations display similar feedback patterns, where accumulated acids like lactic, acetic, or citric acid lower medium pH and, in undissociated form, enter cells causing acidification that disrupts the proton motive force and ATP synthesis. This activates oxidative stress pathways as part of a broader physiological response (Pan et al. 2020).

Exogenous oxidative stress sources include physicochemical stressors arising from bioreactor operation conditions. Physical

and chemical conditions during bioreactor operation serve as major exogenous stressors that induce oxidative stress in microorganisms (García-Caparrós et al. 2021). Notable sources include fluctuating dissolved oxygen levels, mechanical shear forces, and deviations from optimal temperature, pH, and osmotic pressure (Da Rosa et al. 2019). Managing dissolved oxygen is particularly challenging: excessive DO causes hyperoxia and ROS accumulation, while inconsistent mixing and high cell density can generate hypoxic zones followed by damaging reoxygenation cycles (R. Zheng and Pan 2019). Mechanical agitation, while necessary for culture homogeneity, introduces hydrodynamic shear that harms sensitive organisms like filamentous fungi and acts as a biochemical signal increasing ROS via activation of enzymes such as NADH oxidase (Amanullah et al. 2003).

The culture medium in bioreactors can introduce exogenous chemical stressors that directly cause or worsen oxidative stress. These include inhibitory compounds from feedstocks (Alawad and Ibrahim 2024), pro-oxidant byproducts formed during media preparation (Sun et al. 2022), and contaminating metal ions with catalytic activity (Bryukhanov et al. 2022). In food fermentation, complex raw materials and processing-derived compounds can also introduce exogenous redox stress. Plant-based substrates, heat-treated ingredients, phenolic-rich materials, and mineral-containing matrices may contain redox-active compounds or metal ions that alter microbial redox balance and affect fermentation performance (Girio et al. 2017). The harsh pretreatments needed to process these materials can produce inhibitors such as furan aldehydes and phenolics (Kudahettige Nilsson et al. 2016) that disrupt cellular redox balance and increase ROS, impairing microbial performance. Media sterilization often leads to Maillard Reaction Products (MRPs), including Advanced Glycation End-products, which also have pro-oxidant properties (Dai et al. 2024). Industrial-grade substrates may contain trace transition metals like iron and copper that, while beneficial in small amounts, can catalyze reactions generating highly damaging hydroxyl radicals (Bryukhanov et al. 2022).

Microbial interactions in mixed or spontaneous fermentation systems significantly influence oxidative stress levels through complex antagonistic and cooperative mechanisms. Certain lactic acid and acetic acid bacteria produce hydrogen peroxide (H₂O₂), which acts as a form of “chemical warfare” to inhibit sensitive microbes like *Bifidobacterium*, spoilage bacteria, or specific yeasts (Yao et al. 2022). The release of ROS can cause oxidative damage, drive out competitors, and influence the succession and stability of microbial communities—particularly during the early stages of spontaneous fermentations such as kimchi, sauerkraut, kefir, or artisanal cheeses (Mudoor Soorash et al. 2023; Mayo et al. 2021). In contrast, cooperative interactions within these communities help mitigate oxidative stress through mechanisms such as facultative anaerobic yeasts and certain bacteria consuming dissolved oxygen, creating protective niches for strict anaerobes and reducing ROS generation (Boyanova et al. 2024). Community-wide antioxidant mechanisms, like catalase and peroxidase activity, further neutralize ROS and enhance ecosystem resilience (Zandi and Schnug 2022). Therefore, oxidative stress in mixed and spontaneous food fermentations should be interpreted at the community level rather than only at the single-cell level, because

interspecies redox interactions can influence acidification, volatile formation, pigment stability, microbial safety, and fermentation reproducibility.

3 | The Dichotomous Impact of Oxidative Stress

The impact of oxidative stress on microbial fermentation exhibits a fundamental dichotomy that depends critically on the concentration, duration, and spatial distribution of ROS (Figure 2). This dual nature distinguishes between physiological oxidative stress that serves essential regulatory functions and toxicological oxidative stress that causes widespread cellular damage and fermentation inhibition.

Physiological oxidative stress differs from its toxicological form not by the presence of ROS, but by their tightly regulated concentration, dynamics, and cellular management (Sies 2015). In this beneficial state, cells maintain low, balanced ROS levels through a dynamic equilibrium of production and detoxification (Bindoli and Rigobello 2013). This homeostasis prevents harmful buildup and allows ROS to function as important signaling molecules, much like cAMP or Ca^{2+} , rather than indiscriminate agents of damage (Holmström and Finkel 2014; D'Autréaux and Toledano 2007; Sies 2017). As second messengers, ROS enable microorganisms to sense and adapt to changing fermentation conditions. For example, hydrogen peroxide can reversibly oxidize cysteine residues on target proteins, modulating their function and triggering various cellular responses (Bienert and Chaumont 2014; Paulsen and Carroll 2013). These changes affect gene expression, metabolic pathways, cell cycle control, and adaptation to stress (Temple et al. 2005). Common fermentation changes, such as oxygen shifts, nutrient limitations, or pH and temperature fluctuations, often initiate ROS signaling (Grant 2001). For instance, early wort aeration in brewing induces transient ROS generation necessary for yeast to produce sterols and unsaturated fatty acids, supporting membrane integrity and fermentation success (Gibson et al. 2007; Verbelen et al. 2009; Daum et al. 1998).

At low, tightly controlled levels, ROS act as priming cues that precondition microbial cells for subsequent stress, a phenomenon classically termed hormesis or adaptive response (Sies 2017; Davies 2016). Brief, sublethal pulses of H_2O_2 or superoxide initiate redox-dependent switches on transcriptional regulators—such as OxyR, SoxR/SoxS, and PerR in bacteria, and Yap1/Skn7 and Msn2/Msn4 in yeasts and filamentous fungi—thereby launching coordinated programs that elevate cellular resilience beyond the oxidative stress itself (D'Autréaux and Toledano 2007; M. Zheng et al. 1998; Herbig and Helmann 2001; Pomposiello and Demple 2001). The immediate outputs include rapid induction of enzymatic scavengers (SOD, catalase, peroxiredoxins, glutathione peroxidases), reinforcement of thiol-reducing systems (GSH, thioredoxin, glutaredoxin), and activation of protein quality-control modules (chaperones Hsp70/Hsp104, proteases/peptidases) that prevent the accumulation of damaged macromolecules (Imlay 2008; Glover and Lindquist 1998). A hallmark of this redox priming is metabolic reprogramming toward a more reductive intracellular state, where cells transiently reroute carbon through the oxidative pentose phosphate pathway to boost NADPH supply, supporting peroxide detoxification, disulfide reduction, and biosynthesis required for repair (Stincone et al. 2015; Kuehne et al. 2015; Christodoulou et al. 2018).

Crucially, the adaptive program extends beyond protection against the same oxidant to confer cross-protection against heterologous stresses. Cells pre-exposed to mild ROS often exhibit enhanced tolerance to heat, ethanol, organic acids, osmotic upshift, desiccation, or solvent stress (Lewis 2010; Gasch et al. 2000). Mechanistically, this breadth arises because redox-responsive regulators intersect with general-stress networks, mobilizing effectors with wide protective scope including compatible-solute accumulation (e.g., trehalose), chaperone-assisted refolding, membrane fortification, and DNA repair pathways (Gasch et al. 2000; Elbein 2003; Friedberg 2003).

In food fermentation systems, such redox priming and cross-protection are particularly relevant to starter culture performance and process stability. For example, *S. cerevisiae*

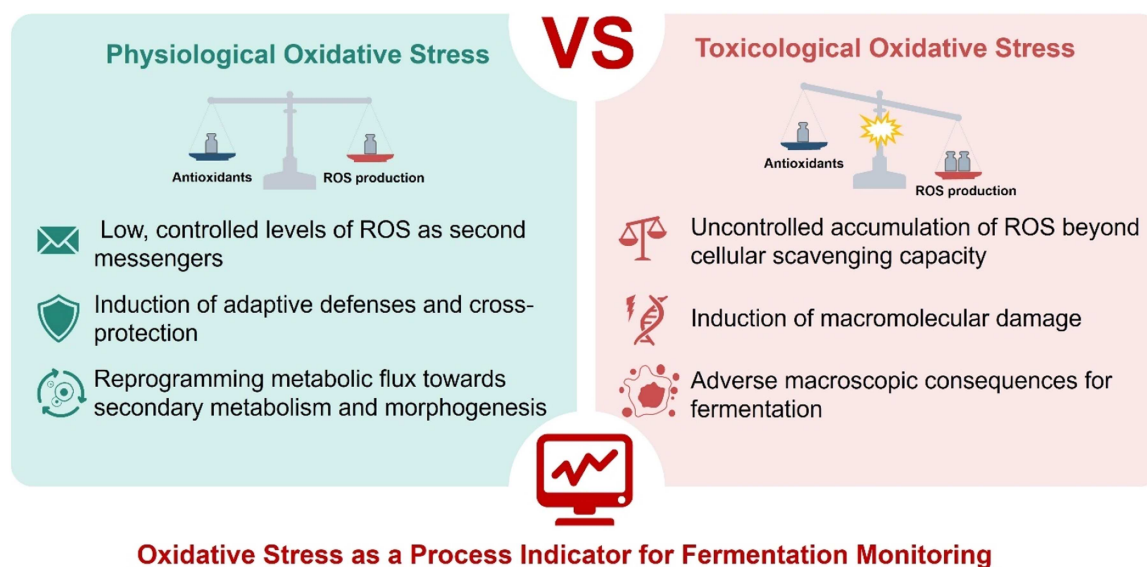


FIGURE 2 | The dichotomous impact of oxidative stress.

used in brewing and winemaking may encounter combined oxidative, ethanol, osmotic, and nutrient stresses, and its redox adaptation can influence fermentation completion, aroma formation, and yeast survival. Similarly, lactic acid bacteria in dairy and vegetable fermentations may experience oxidative pressure during oxygen exposure, storage, or mixed-culture succession, which can affect acidification capacity, probiotic viability, and product stability.

Physiological oxidative stress drives characteristic reallocation of carbon and reducing power that extends well beyond immediate detoxification. Low, spatially confined pulses of ROS trigger redox-sensitive regulators to reprioritize flux from growth-associated biosynthesis toward protective and differentiation programs (Han et al. 2001). A rapid consequence is rerouting of glucose-6-phosphate into the oxidative pentose phosphate pathway, elevating NADPH to support peroxide removal, disulfide reduction, Fe-S cluster repair, and reductive biosynthesis (Stincone et al. 2015; Christodoulou et al. 2018; Krüger et al. 2011). Within this reprogrammed landscape, secondary metabolism is selectively favored. Many polyketide, nonribosomal peptide, terpenoid, and pigment pathways are transcriptionally or post-translationally responsive to redox cues and are intensely NADPH- and acetyl-/malonyl-CoA-demanding (Brakhage 2013; Keller 2019). In *Streptomyces* and filamentous fungi, this often coincides with developmental milestones (aerial mycelium/conidiation), when secondary metabolite synthesis peaks (Gessler et al. 2007; Bibb 2005). In filamentous fungi, ROS act locally at hyphal tips together with Ca^{2+} and small GTPases to coordinate polarity, branching, and septation (Verbelen et al. 2009; Olagunju 2008; Reverberi et al. 2006).

In stark contrast, toxicological oxidative stress arises when the production rate of ROS persistently exceeds the cellular capacity for detoxification and repair. Unlike the localized, pulsed signals characteristic of physiological “eustress” this distress state is defined by sustained, high intracellular concentrations of ROS, prolonged exposure that outlasts adaptive transcriptional responses, and loss of spatial control with diffuse propagation of oxidants across compartments (Sies 2017; Imlay 2013; Danilewicz 2012). Functionally, the antioxidant buffering network becomes saturated: the GSH/GSSG ratio collapses, NADPH pools are depleted, and key scavengers enter inactive states such as peroxiredoxin hyperoxidation to sulfinic/sulfonic forms and catalase/SOD inhibition (Christodoulou et al. 2018; Netto and Antunes 2016; Jones 2006; Rhee and Woo 2011). Concomitantly, redox potential shifts toward oxidizing values (elevated ORP/Eh), reflecting systemic failure of redox homeostasis (Danilewicz 2012; Jones 2008).

At the chemical level, uncontrolled ROS sets off autocatalytic amplification loops where hydrogen peroxide accumulates and, in the presence of free or loosely chelated transition metals, fuels Fenton/Haber-Weiss chemistry that generates hydroxyl radicals with near-diffusion-limited reactivity (Imlay 2013; Halliwell and Gutteridge 2015; Winterbourn 1995). Iron-sulfur enzymes such as aconitase and dehydratases are rapidly inactivated, throttling central carbon flux and respiratory throughput (Imlay 2013; Gardner and Fridovich 1991). The biochemical injury pattern is broad and self-reinforcing, involving lipid peroxidation that initiates chain reactions

compromising membrane integrity, protein oxidation including carbonylation and disulfide scrambling that disables enzymes and chaperones, and nucleic-acid damage producing 8-oxo-dG and strand breaks that trigger checkpoint responses and replication stress (Halliwell and Gutteridge 2015; Cooke et al. 2003; Dalle-Donne et al. 2006; Cadet and Wagner 2013). Hallmark biomarker shifts include elevated malondialdehyde (MDA), protein carbonyls, and intracellular ROS indicators, together with diminished antioxidant enzyme activities and thiol pools (Dalle-Donne et al. 2006).

From a food fermentation perspective, the importance of these oxidative stress mechanisms lies not only in their molecular architecture, but also in their effects on fermentation performance and product quality. Redox-sensitive regulation can alter microbial growth, acidification capacity, alcohol tolerance, enzyme secretion, flavor-active metabolite production, and survival during storage. Therefore, oxidative stress pathways should be interpreted in relation to specific microorganisms, substrates, fermentation stages, and quality endpoints, rather than as isolated stress-response circuits.

4 | Cellular Detection and Response Mechanisms

Microorganisms have evolved sophisticated molecular systems to detect and respond to oxidative stress, enabling rapid adaptation to changing redox conditions during fermentation through complex sensing networks and multilayered defense mechanisms that orchestrate coordinated cellular responses to maintain homeostasis under challenging oxidative conditions.

To clarify figure organization and avoid presenting these diagrams as generic pathway summaries, Figures 3–10 summarize representative bacterial and fungal redox regulatory modules discussed in this section. These figures are intended to facilitate comparison among major oxidative stress sensing systems relevant to fermentation microorganisms, rather than to provide exhaustive pathway maps for all microbial species.

In bacterial systems, the OxyR transcriptional regulator serves as the primary hydrogen peroxide sensor, undergoing reversible cysteine oxidation upon H_2O_2 exposure that leads to conformational changes activating transcription of antioxidant genes including catalase (*katG*), alkyl hydroperoxide reductase (*ahpCF*), and glutaredoxin (*grxA*) (Figure 3) (M. Zheng et al. 1998; Carmel-Harel and Storz 2000). The SoxR/SoxS regulatory cascade provides complementary detection of superoxide and nitric oxide stress through SoxR, which contains a [2Fe-2S] cluster that undergoes oxidation upon exposure to superoxide, activating transcription of *soxS* (Figure 4) (Pomposiello and Demple 2001). SoxS then induces expression of antioxidant enzymes including superoxide dismutase (*sodA*), glucose-6-phosphate dehydrogenase (*zwf*), and DNA repair proteins, creating a comprehensive defense network. In Gram-positive bacteria, PerR acts as an iron-dependent peroxide sensor that binds to promoter regions and represses antioxidant gene expression under normal conditions, but upon peroxide exposure, iron-catalyzed oxidation of PerR leads to its inactivation and derepression of target genes (Figure 5) (Herbig and Helmann 2001). The general stress response is coordinated

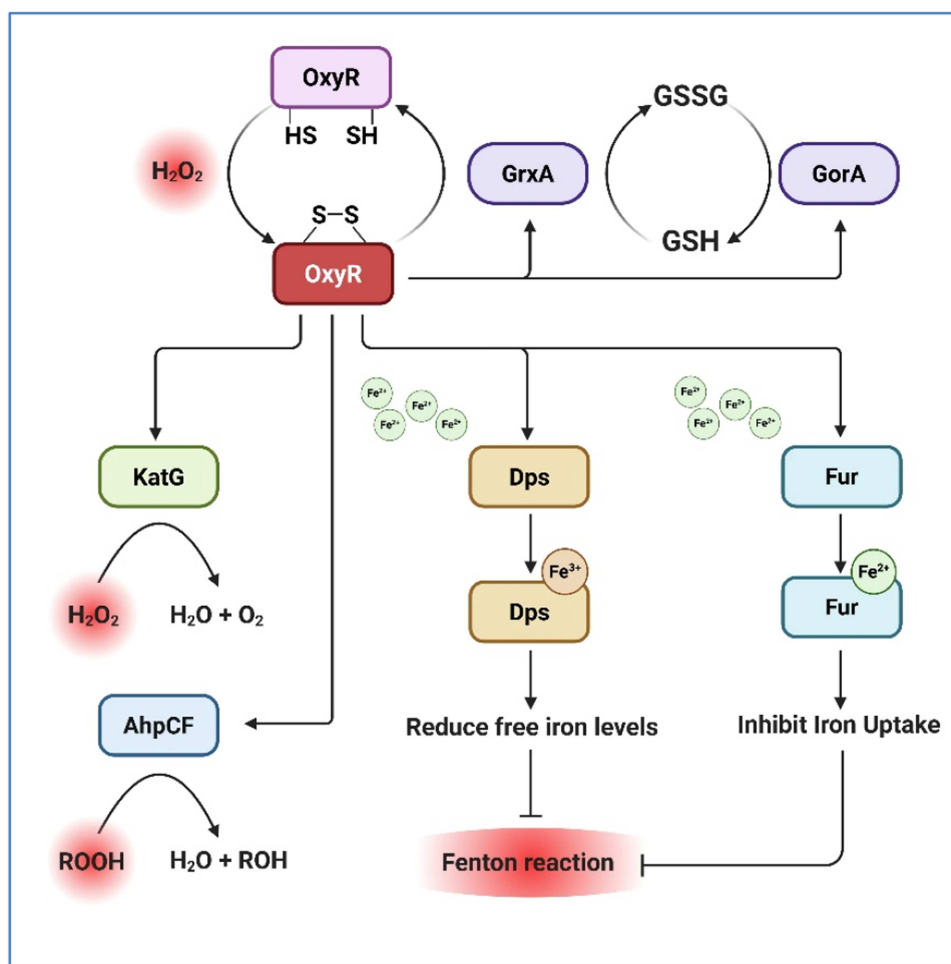


FIGURE 3 | Mechanism of the OxyR regulon. OxyR-mediated hydrogen peroxide sensing in bacteria. This diagram summarizes how OxyR detects H_2O_2 through reversible cysteine oxidation and activates antioxidant genes such as *katG*, *ahpCF*, and *grxA*, thereby supporting peroxide detoxification and bacterial survival under oxidative stress.

by RpoS ($\sigma 38$), which integrates multiple stress signals including oxidative stress and controls expression of genes involved in stationary phase survival, stress tolerance, and metabolic adjustments that enhance cellular resilience under adverse conditions (Figure 6) (Hengge-Aronis 2002; Battesti et al. 2011). The IscR regulon plays a crucial role in iron-sulfur cluster biogenesis and repair, responding to oxidative damage of Fe-S clusters by derepressing genes involved in cluster assembly and iron homeostasis, thereby maintaining the activity of essential metalloproteins under oxidative stress conditions (Figure 7) (Schwartz et al. 2001).

Together, Figures 3–7 highlight bacterial modules that respond to peroxide stress, superoxide or redox-active stress, general stress, and Fe-S cluster damage, thereby linking oxidative stress perception with antioxidant defense, redox homeostasis, and survival under fermentation-related stress conditions.

Fungal systems utilize different but equally sophisticated detection mechanisms that integrate oxidative stress responses with broader cellular stress networks. In yeasts, Yap1 serves as the master regulator of oxidative stress response, undergoing disulfide bond formation under oxidizing conditions that promotes nuclear accumulation and transcriptional activation of antioxidant genes

including catalase, glutathione peroxidase, and thioredoxin (Figure 8). Skn7 works in parallel as a response regulator in a two-component signaling system, responding to both osmotic and oxidative stress through phosphorylation-dependent activation that regulates expression of genes involved in cell wall integrity and oxidative stress tolerance. The High Osmolarity Glycerol (HOG-MAPK) pathway integrates osmotic and oxidative stress signals through a sophisticated kinase cascade that leads to activation of stress-responsive transcription factors and metabolic adjustments, including glycerol accumulation and cell cycle arrest, that enhance cellular survival under stress conditions (Figure 9) (Hohmann 2002). Msn2/Msn4 transcription factors coordinate the general stress response in yeasts, integrating signals from multiple stress conditions including oxidative stress, heat shock, and nutrient limitation, and regulating expression of genes involved in stress tolerance, carbohydrate metabolism, and cell cycle control through binding to stress response elements (STRE) in target gene promoters (Figure 10).

Figures 8–10 summarize fungal oxidative stress response modules that are particularly relevant to yeasts and filamentous fungi used in brewing, winemaking, sourdough, and mold-based fermentations, where redox regulation can affect stress tolerance, growth, enzyme secretion, and fermentation performance.

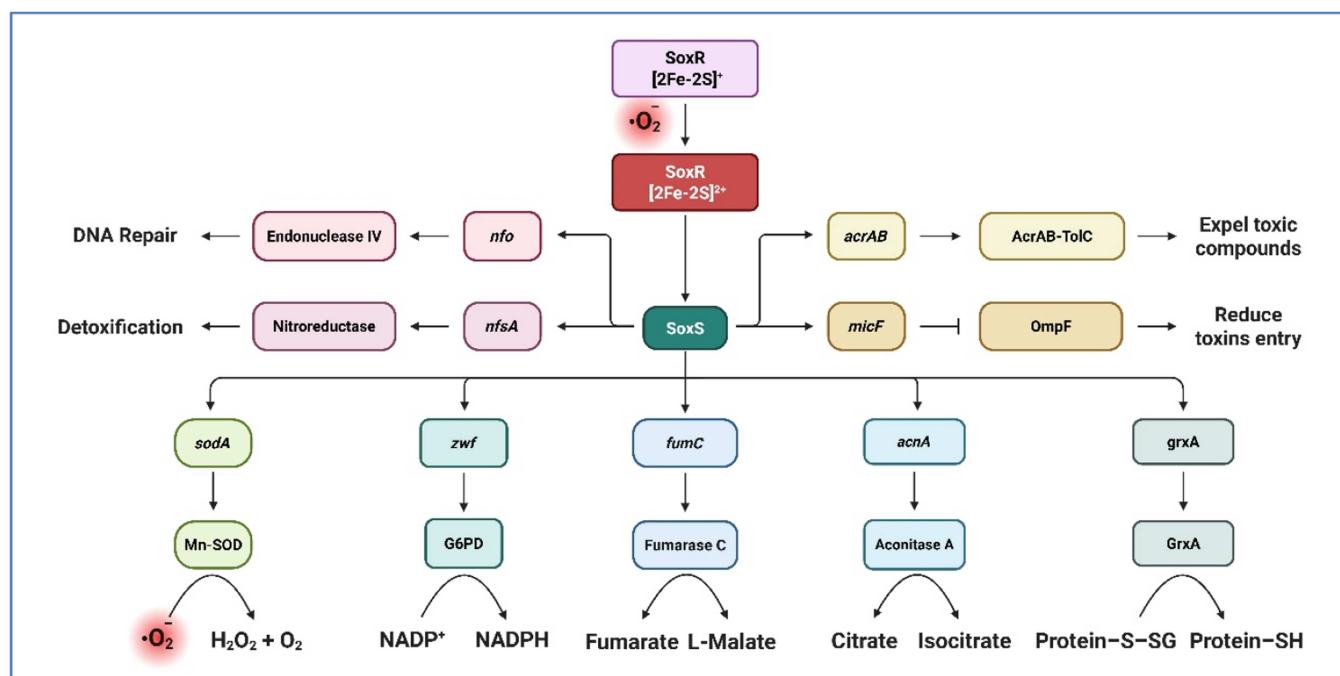


FIGURE 4 | Mechanism of the SoxRS two-component system. SoxRS-mediated response to superoxide and redox-active stress in bacteria. SoxR senses oxidative stress through its [2Fe-2S] cluster and activates SoxS, which regulates genes involved in superoxide detoxification, NADPH supply, DNA repair, and redox homeostasis.

Taken together, oxidative stress regulatory systems should be interpreted according to the microbial group and fermentation context in which they operate. In bacterial fermentations, regulators such as OxyR, SoxRS, PerR, RpoS, and IscR are particularly relevant to lactic acid bacteria, acetic acid bacteria, and *Bacillus*-dominated systems, where redox responses may influence oxygen tolerance, acidification performance, microbial succession, oxidative survival, and safety-related stability. In yeasts used for brewing and winemaking, redox regulation contributes to ethanol tolerance, membrane stability, fermentation completion, and aroma metabolism. In filamentous fungi involved in soy-based and mold-fermented foods, oxidative stress responses may affect aerobic growth, enzyme secretion, sporulation, secondary metabolism, and flavor precursor formation. In mixed communities, the overall redox state further reflects interspecies oxygen consumption, ROS production, antioxidant enzyme activity, and niche construction rather than the response of a single organism. These organism- and process-dependent differences connect oxidative stress perception with fermentation performance and product quality.

Following oxidative stress detection, microorganisms activate multilayered antioxidant defense networks that combine enzymatic scavenging, nonenzymatic redox buffering, and damage repair. Enzymatic systems form the first line of defense: SODs convert superoxide anions into H_2O_2 and oxygen through isoforms with different metal cofactors and cellular localizations (Fridovich 1995), catalase rapidly decomposes H_2O_2 into water and oxygen, and peroxiredoxins reduce H_2O_2 and organic hydroperoxides through thioredoxin- or glutaredoxin-dependent reactions. Nonenzymatic antioxidants further support redox homeostasis, among which glutathione (GSH) acts as a major cellular reductant and ROS scavenger; the GSH/GSSG ratio is therefore widely used as an indicator of intracellular redox state

and early oxidative stress (Jones 2006). When oxidative damage escapes these buffering systems, repair and maintenance pathways, including methionine sulfoxide reductases, protein disulfide isomerases, Hsp70/Hsp104 chaperone systems, and DNA repair pathways, help restore protein function and maintain genome integrity (Glover and Lindquist 1998). In food fermentation microorganisms, the efficiency of these defense layers directly affects starter culture survival, stress tolerance, fermentation continuity, and product stability.

From a comparative perspective, several oxidative stress responses are broadly conserved across fermentation microorganisms, including ROS scavenging by SOD, catalase or peroxidase systems, thiol-based redox buffering, NADPH-dependent repair, and protein quality-control mechanisms. In contrast, the upstream sensing and regulatory architectures are more organism-specific: OxyR, SoxRS, PerR, RpoS, and IscR are mainly associated with bacterial responses, whereas Yap1/Skn7, HOG-MAPK, and Msn2/Msn4 are more representative of yeasts and filamentous fungi. This distinction is important because conserved defense modules may support general stress tolerance, whereas organism-specific regulatory systems determine how different starters or microbial communities respond to oxygen exposure, matrix-derived oxidants, and fermentation-stage-specific stress.

5 | Detection and Monitoring Strategies

Effective management of oxidative stress in fermentation systems requires comprehensive detection and monitoring strategies that combine traditional biomarker analysis with advanced online monitoring and multi-omics technologies to provide both real-time process information and detailed biochemical insights into cellular redox status (Figure 11). Biomarker-based detection forms

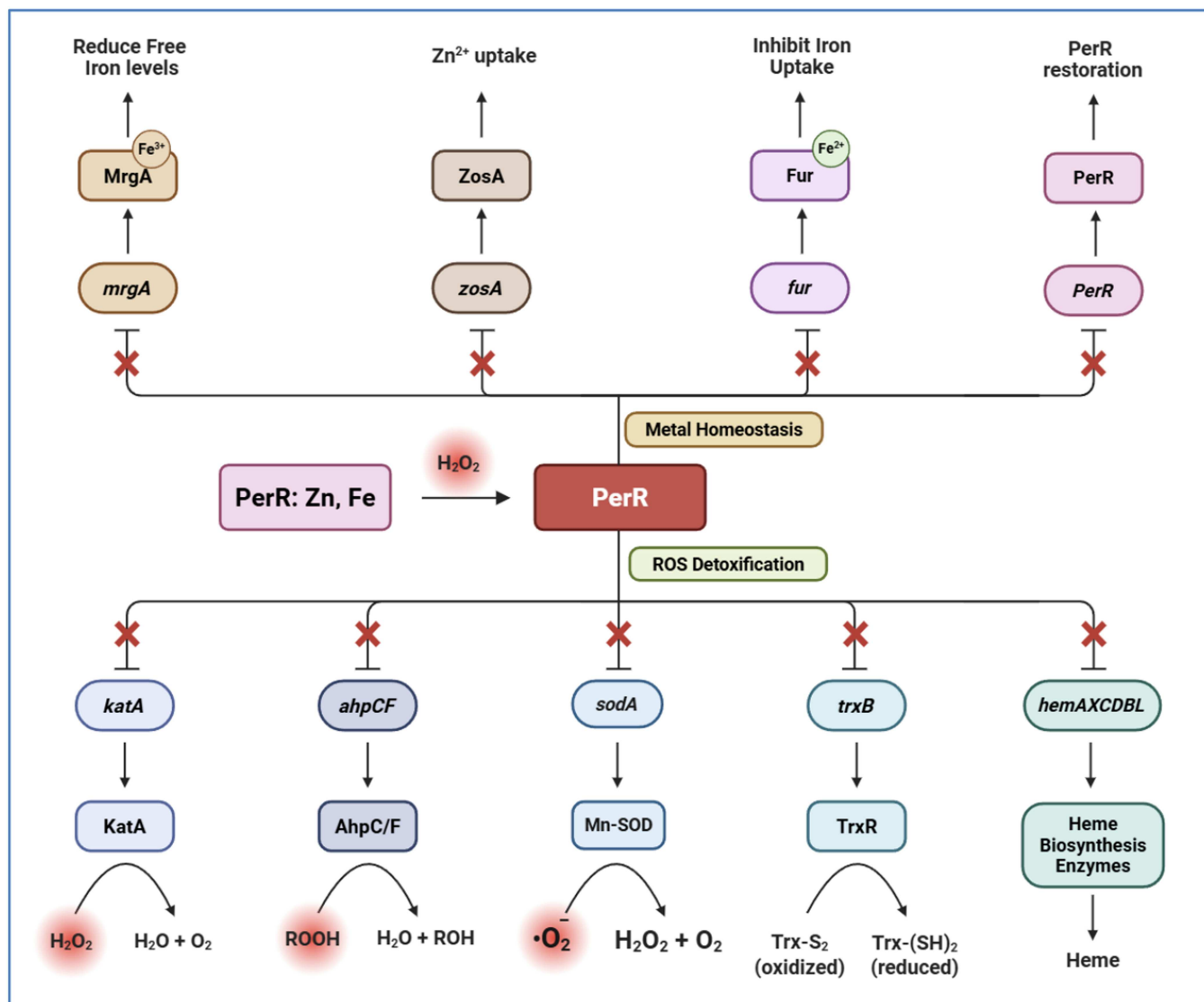


FIGURE 5 | Mechanism of the PerR regulon. PerR-mediated peroxide response in Gram-positive bacteria. Under nonstress conditions, PerR represses peroxide defense genes. Upon peroxide exposure, metal-catalyzed oxidation inactivates PerR and derepresses antioxidant genes, supporting bacterial adaptation to oxidative stress.

the foundation of oxidative stress assessment, utilizing both direct ROS measurement and oxidative damage markers to quantify the extent and nature of oxidative stress. Fluorescent probe technologies have revolutionized direct ROS detection, with DCFH-DA (2',7'-dichlorofluorescein diacetate) serving as a general ROS detection probe that becomes fluorescent upon oxidation and is suitable for flow cytometry and microscopy applications. More specific probes include DHE (Dihydroethidium) for superoxide-specific detection that forms red fluorescent products upon reaction with $O_2^{\bullet-}$, Amplex Red for hydrogen peroxide quantification often used in combination with horseradish peroxidase, and MitoSOX for mitochondrial superoxide measurement that provides subcellular localization of ROS production. Electrochemical methods using specialized electrodes provide real-time monitoring capabilities with high sensitivity that can be integrated into process control systems for continuous assessment of oxidative stress levels. Because quantitative relationships between redox indicators and fermentation outcomes vary among microorganisms, matrices, and processing stages, ORP, DO, ROS biomarkers, and

antioxidant capacity should be calibrated against process-specific endpoints, such as acidification rate, ethanol production, microbial viability, volatile formation, color stability, texture retention, and shelf life, before being used as industrial decision thresholds.

Oxidative damage markers provide crucial information about the extent and nature of cellular injury resulting from oxidative stress. Lipid peroxidation is assessed through MDA measured using the TBARS (thiobarbituric acid reactive substances) assay that indicates membrane damage severity, 4-hydroxynonenal (4-HNE) detected through immunoassays or mass spectrometry as a key marker of lipid peroxidation, and isoprostanes measured by mass spectrometry for precise oxidative damage assessment. Protein oxidation markers include protein carbonyls measured using DNPH (2,4-dinitrophenylhydrazine) derivatization representing irreversible protein oxidation, advanced oxidation protein products as spectrophotometric markers of protein oxidation, and nitrotyrosine indicating peroxynitrite-mediated protein damage. DNA damage is evaluated through 8-oxo-deoxyguanosine (8-oxo-dG)

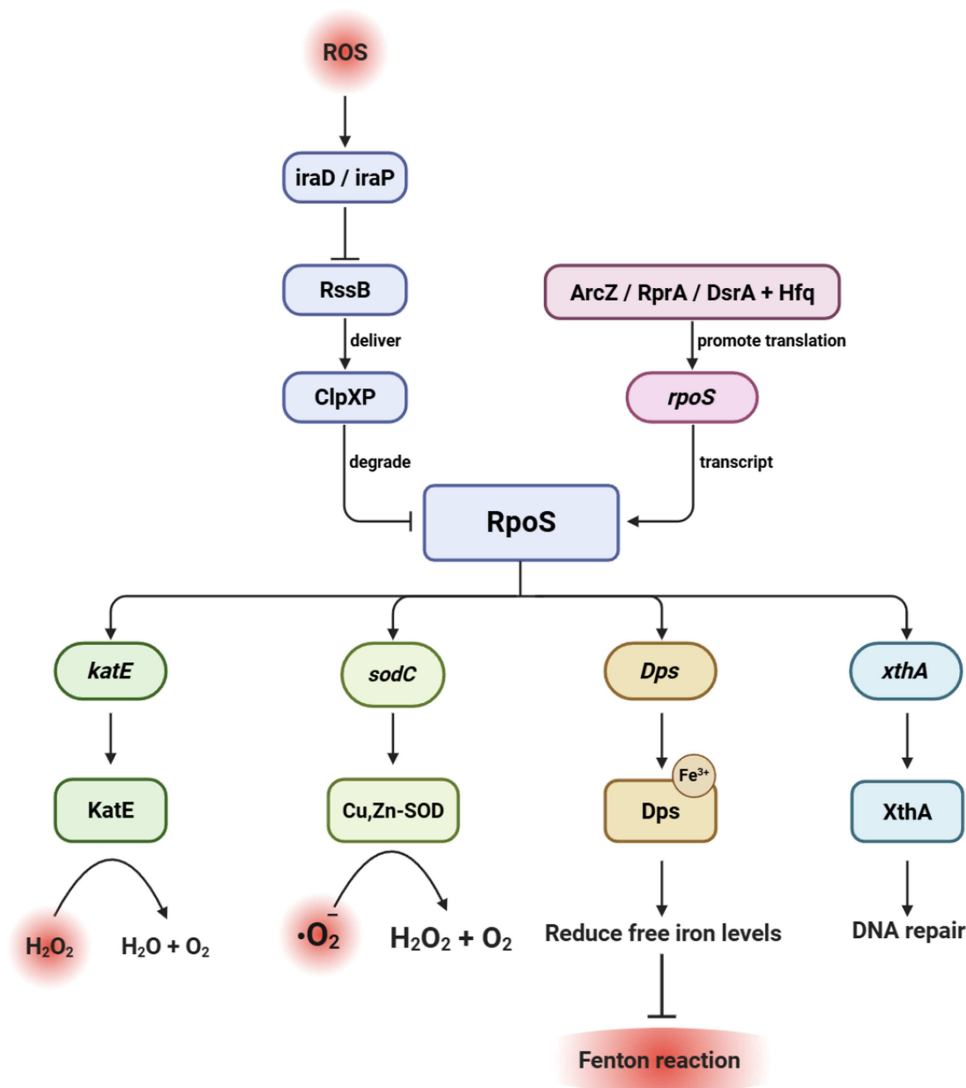


FIGURE 6 | Mechanism of the RpoS (σ_S) regulon. RpoS-mediated general stress response in bacteria. RpoS integrates oxidative stress with other environmental stresses and regulates genes involved in stationary-phase survival, stress tolerance, metabolic adjustment, and long-term cellular resilience.

measured by HPLC–MS/MS as the most common oxidative DNA lesion, DNA strand breaks assessed using the comet assay for single-cell gel electrophoresis, and micronucleus formation as a cytogenetic endpoint for chromosomal damage assessment.

Process monitoring parameters provide real-time insights into fermentation redox status and enable proactive control strategies through online measurement systems. Online redox potential (ORP) measurement offers immediate feedback on overall redox conditions, with normal fermentation typically showing ORP values from -200 to -400 mV indicating a reducing environment, while oxidative stress conditions shift ORP toward positive values signaling oxidizing conditions. Implementation requires careful electrode selection and calibration protocols, temperature and pH compensation, and integration with automated control systems for effective process guidance. Dissolved oxygen dynamics reveal critical information about oxidative stress development through oxygen uptake rate indicating metabolic activity and potential respiratory stress, critical dissolved oxygen concentration (DOcrit) as the threshold for oxygen limitation, and oxygen

transfer coefficient (kLa) measuring mass transfer efficiency and mixing effectiveness. Advanced monitoring approaches include multipoint DO measurement for spatial heterogeneity assessment, dynamic DO profiling during different fermentation phases, and correlation with other process parameters for comprehensive analysis. Exhaust gas analysis provides indirect but valuable metabolic information through respiratory quotient indicating metabolic state as the ratio of CO_2 production to O_2 consumption, carbon dioxide evolution rate reflecting overall metabolic activity, and oxygen consumption rate indicating respiratory chain activity and potential electron leakage.

Multi-omics approaches provide systems-level insight into oxidative stress responses by linking gene expression, protein abundance, redox-related metabolites, and metabolic flux changes. Transcriptomics can identify stress-responsive genes involved in antioxidant defense, heat shock response, DNA repair, and metabolic adjustment, including genes encoding antioxidant enzymes, chaperones, and repair proteins. Proteomics and post-translational modification analyses further reveal

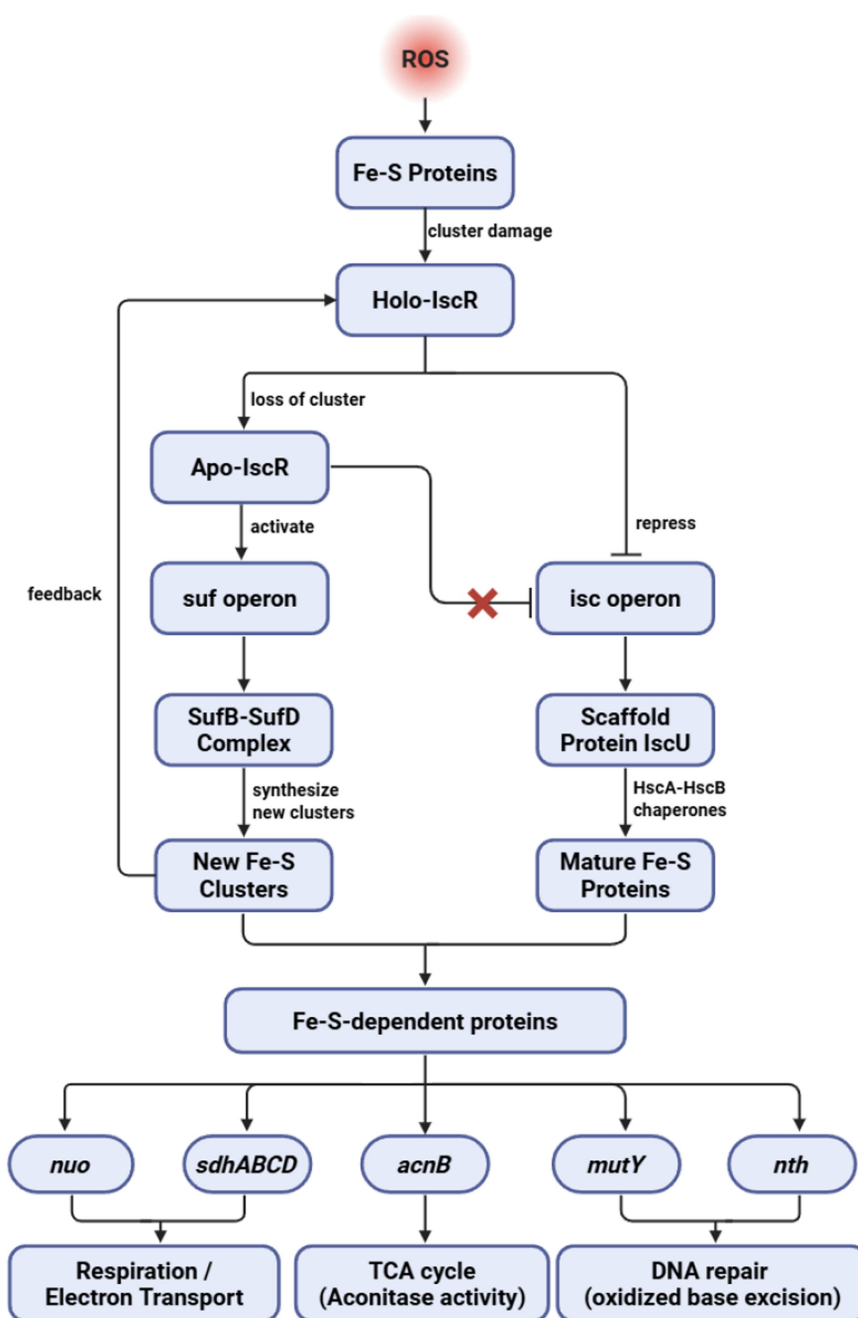


FIGURE 7 | Mechanism of the IscR regulon. IscR-mediated regulation of iron–sulfur cluster homeostasis. IscR responds to oxidative damage of Fe–S clusters and coordinates genes involved in Fe–S cluster biogenesis, repair, and iron homeostasis, thereby maintaining essential metalloprotein function during oxidative stress.

changes in antioxidant enzyme abundance, stress protein expression, redox-sensitive protein complexes, protein carbonylation, S-glutathionylation, nitrosylation, and nitration. Metabolomics complements these data sets by quantifying redox cofactors such as NADH/NAD⁺, NADPH/NADP⁺, GSH/GSSG, and ATP/ADP, as well as antioxidant metabolites, oxidative damage products, and pathway intermediates. Metabolic flux analysis further helps identify the redistribution of carbon and reducing power through pathways such as the pentose phosphate pathway, TCA cycle, and amino acid metabolism. For food fermentation applications, these omics-derived markers should be interpreted together with process parameters and quality endpoints rather than used as standalone indicators.

Monitoring approaches also differ in their practical readiness. Fluorescent probes, intracellular ROS assays, flow cytometry, microscopy, and omics-based analyses are valuable for laboratory research because they provide mechanistic and cell-level information, but they are often difficult to implement directly in industrial fermentations. In contrast, ORP, DO, pH, temperature, gas analysis, and selected spectroscopic or electrochemical sensors are more feasible for pilot- or industrial-scale monitoring because they can be integrated into online or at-line process control. Therefore, laboratory biomarkers should be used to identify mechanistic links, whereas industrial monitoring should prioritize robust, noninvasive, calibratable, and matrix-tolerant indicators that can be linked to fermentation performance and product quality.

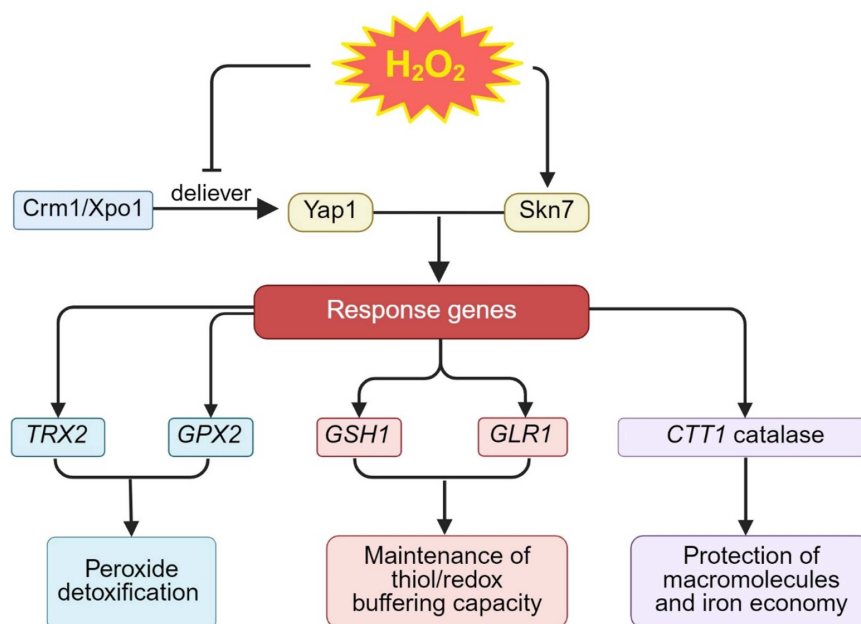


FIGURE 8 | Mechanism of the Yap1/Skn7 pathway. Yap1/Skn7-mediated oxidative stress response in yeasts and fungi. Yap1 and Skn7 regulate antioxidant gene expression through redox-sensitive and stress-responsive signaling, contributing to peroxide detoxification, thiol redox balance, and fungal stress tolerance.

Despite these advances, oxidative stress monitoring in real food fermentation matrices remains challenging. Complex substrates rich in pigments, polyphenols, proteins, lipids, salts, and suspended particles may interfere with fluorescent probes, spectrophotometric assays, electrochemical sensors, and omics-based measurements. Common ROS probes may lack absolute specificity and can respond to multiple oxidants or redox-active food components. It is also difficult to distinguish intracellular ROS from extracellular oxidants and matrix-derived pro-oxidants in mixed fermentations. Sampling procedures may alter oxygen exposure and redox status, while ORP and DO electrodes require careful calibration, cleaning, and compensation for pH, temperature, fouling, and matrix effects. Therefore, laboratory biomarkers should be validated against process-specific performance and quality endpoints before being used as real-time industrial decision parameters.

In phenolic-rich plant matrices, antioxidant capacity and redox-active compounds may further complicate the interpretation of ROS probes and oxidative damage markers, because matrix-derived antioxidants can quench radicals, interfere with colorimetric or fluorescent signals, and decouple measured antioxidant capacity from intracellular oxidative stress (Zhang et al. 2023).

6 | Mitigation and Control Strategies

In practical food fermentation, oxidative stress management can be organized into three complementary levels. The first level is prevention, which aims to reduce excessive ROS generation through oxygen management, feeding optimization, temperature and pH control, and minimization of pro-oxidant compounds in raw materials. The second level is reinforcement, which enhances microbial robustness through strain selection, adaptive evolution, starter culture protection, or targeted

improvement of antioxidant capacity. The third level is real-time monitoring-based adaptive regulation, which integrates real-time indicators such as ORP, DO, selected oxidative stress biomarkers, and product quality endpoints to guide process adjustment. The applicability of each strategy depends on the fermentation type, dominant microorganisms, oxygen requirement, food matrix, regulatory constraints, and production scale.

Within this framework, process engineering mainly contributes to the prevention of excessive ROS generation. Oxygen management is particularly important for aerobic and microaerobic fermentations, but the optimal oxygen window is highly process-specific. For some aerobic or microaerobic processes, maintaining DO within an empirically optimized range can support microbial growth while limiting excessive ROS formation. However, this range should be validated for each microorganism, substrate, fermentation stage, and target product quality rather than being treated as a universal threshold. Cascade control systems, micro-aeration, phase-dependent aeration, and feedback regulation based on ORP, DO, or selected stress biomarkers may therefore be used as process-specific tools rather than general prescriptions.

Substrate feeding and medium optimization further contribute to the prevention level by reducing metabolic overflow and pro-oxidant accumulation. Fed-batch cultivation can prevent substrate overflow metabolism and associated redox imbalance, whereas exponential feeding better matches substrate supply to cellular growth rate and metabolic capacity. Multisubstrate feeding may also help balance carbon and nitrogen metabolism, thereby reducing redox stress caused by excessive carbon flux. In addition, gradual substrate introduction, coordinated carbon and nitrogen delivery, pH-stat feeding, optimized temperature trajectories, and improved mixing design can reduce metabolic

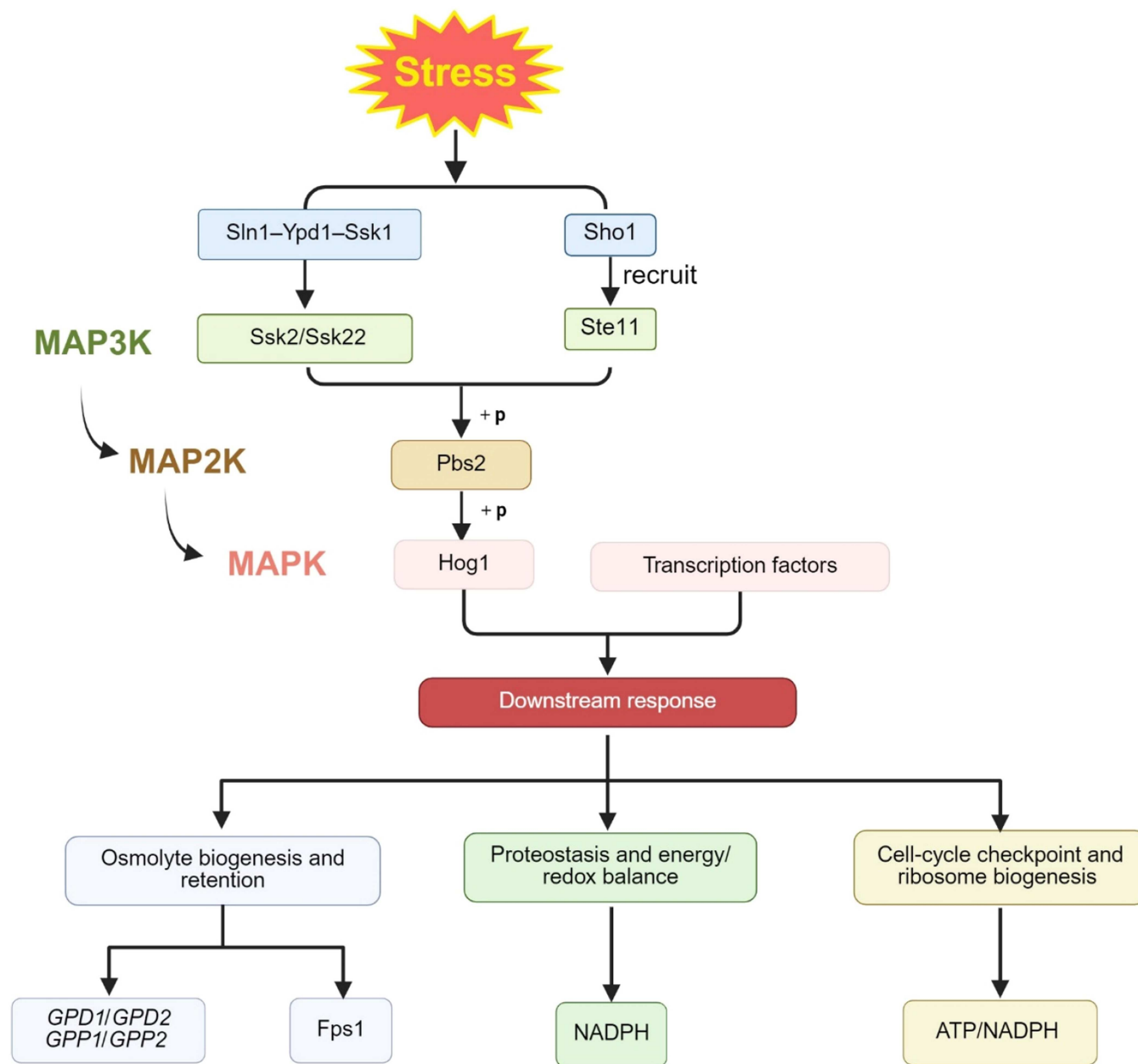


FIGURE 9 | Mechanism of the HOG-MAPK Pathway. HOG-MAPK pathway in fungal stress adaptation. The HOG-MAPK cascade integrates osmotic and oxidative stress signals and regulates downstream transcriptional and metabolic responses, including glycerol accumulation, cell cycle control, and stress adaptation.

shock, oxygen heterogeneity, and shear-associated oxidative stress.

At the reinforcement level, strain selection and strain improvement aim to enhance microbial tolerance to oxidative stress. This can be achieved by selecting robust starter cultures, using adaptive evolution, or improving endogenous antioxidant capacity. Genetic or metabolic engineering strategies may include enhancing SOD, catalase, peroxiredoxin, glutathione peroxidase, thioredoxin, glutaredoxin, or NADPH-generating pathways. However, the applicability of engineered strains in food fermentation depends on food safety assessment, regulatory approval, genetic stability, and consumer acceptance. Nevertheless, strain-based strategies should be evaluated case by case in food fermentation systems. Although strain

engineering can enhance oxidative stress tolerance, its practical use depends on food safety assessment, genetic stability, regulatory approval, labeling requirements, and consumer acceptance. Therefore, engineered or improved strains should be validated not only for stress tolerance, but also for fermentation performance, sensory quality, and product safety.

Stress response optimization through transcriptional engineering involves enhancing the activity of redox-responsive regulators such as OxyR, SoxR, and Yap1, improving general stress tolerance through global regulator modification, and tuning temporal responses for faster activation and appropriate duration of stress responses. Metabolic robustness engineering focuses on central metabolism reinforcement to enhance flux capacity through glycolysis and the TCA cycle, overflow

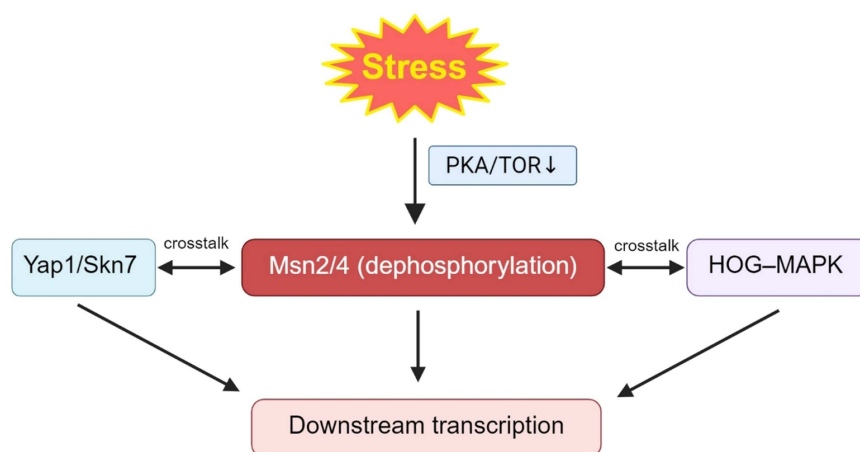


FIGURE 10 | Mechanism of the Msn2/Msn4 transcription factors. Msn2/Msn4-mediated general stress response in yeasts. Msn2/Msn4 transcription factors bind stress response elements and regulate genes involved in oxidative stress tolerance, carbohydrate metabolism, heat shock response, and survival under multiple fermentation-related stresses.

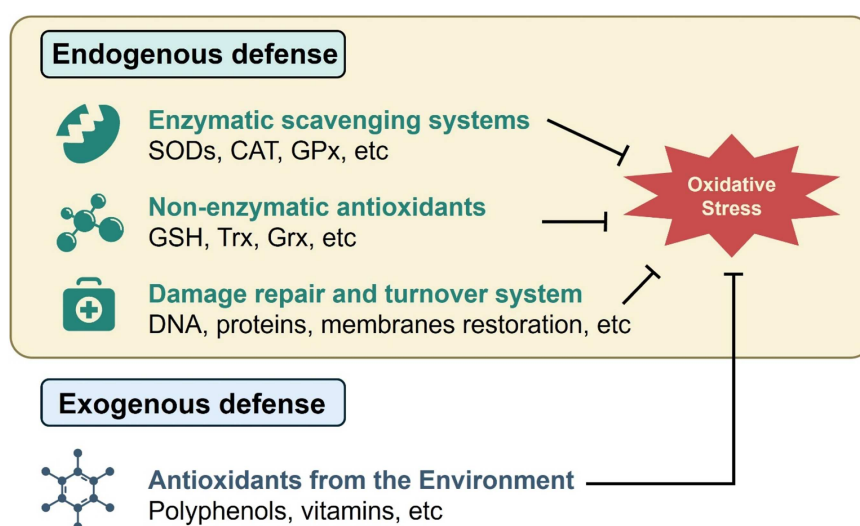


FIGURE 11 | Core microbial strategies for managing oxidative stress.

metabolism prevention to reduce acetate and other organic acid formation that can contribute to stress, and energy metabolism optimization for improved ATP yield and efficiency. Membrane engineering approaches include lipid composition modification for enhanced membrane stability under oxidative conditions, sterol content optimization for improved membrane integrity, and transport system enhancement for better nutrient uptake and waste product efflux.

Medium formulation acts at both the prevention and reinforcement levels by reducing pro-oxidant inputs and supporting endogenous antioxidant capacity. Food-grade antioxidants, antioxidant precursors, vitamins, and metal chelators may help reduce oxidative damage when permitted by relevant regulations and when their sensory effects are acceptable. Meanwhile, pro-oxidant minimization can be achieved through raw material screening, removal of excessive transition metals or inhibitory compounds, optimization of sterilization conditions, and appropriate storage of medium components. These measures should be evaluated together with microbial

performance and product quality rather than applied as universal interventions.

Pro-oxidant minimization is equally important and involves comprehensive raw material treatment strategies. Detoxification processes using activated charcoal treatment, ion exchange, and other purification methods remove harmful compounds that could contribute to oxidative stress, while enzymatic treatments with laccase and peroxidase eliminate phenolic compounds that can disrupt cellular redox balance. Physical separation methods including filtration and centrifugation remove contaminants that could catalyze ROS formation. Media preparation optimization focuses on sterilization condition optimization using reduced time-temperature combinations to minimize MRPs, pH adjustment strategies that minimize pro-oxidant formation during media preparation, and storage condition control to prevent oxidative deterioration of media components. Quality control measures ensure consistent oxidative stress management through raw material screening for pro-oxidant compounds and metal contaminants, media

analysis to monitor antioxidant capacity and pro-oxidant levels, and batch-to-batch consistency protocols that ensure reproducible oxidative stress conditions throughout production campaigns.

These management strategies should therefore be evaluated by their effects on measurable fermentation outcomes, including microbial viability, fermentation kinetics, metabolite production, sensory quality, storage stability, and safety-related indicators.

Overall, oxidative stress mitigation strategies should be selected according to the fermentation type, dominant microorganisms, oxygen requirement, substrate composition, regulatory constraints, process scale, and target quality attributes, rather than being applied as universal interventions across all fermented foods. Accordingly, oxidative stress control should rely on validated process-specific redox windows, and these windows should be linked to measurable fermentation performance and product quality outcomes before being used to guide process decisions.

7 | Applications in Food Fermentation

From a food quality perspective, oxidative stress affects fermentation outcomes through both microbial and matrix-dependent routes. At the microbial level, redox imbalance can influence cell viability, stress tolerance, metabolic activity, acidification capacity, and the production of flavor-active metabolites. At the food matrix level, oxidative reactions may affect lipid oxidation, pigment stability, protein functionality, aroma retention, texture development, and off-flavor formation. Therefore, oxidative stress in fermented foods should be evaluated not only by microbial survival or ROS accumulation, but also by sensory quality, nutritional value, storage stability, and shelf life.

These effects are expressed differently across food categories. In alcoholic beverages, oxidative stress may influence yeast viability, fermentation completion, aroma metabolism, and oxidative off-flavors. In dairy fermentations, it may affect starter activity, probiotic survival, lipid oxidation, texture development, and storage stability. In vegetable and plant-based fermentations, it may influence pigment preservation, tissue softening, acidification stability, phenolic transformation, aroma retention, and shelf life.

To provide a more structured comparison across food fermentation systems, Table 1 summarizes representative fermentation types, key microorganisms, major oxidative stress sources, commonly used biomarkers or indicators, process or quality impacts, and reported or potential control strategies.

The principles of oxidative stress management find practical application across diverse food fermentation systems, where controlling ROS levels is essential for product quality, process efficiency, and microbial health, demonstrating how understanding oxidative stress mechanisms translates into improved fermentation outcomes and product characteristics. Alcoholic beverage production represents one of the most extensively

studied applications of oxidative stress management in fermentation, with beer production involving multiple stages where oxidative stress control is critical. Wort aeration optimization requires controlled oxygen levels of 8–12 ppm to provide adequate oxygen for yeast growth without excessive ROS formation. The natural antioxidant compounds in hops, including polyphenols and iso- α -acids, provide protection against oxidation throughout the brewing process, while yeast strain selection focuses on varieties with enhanced oxidative stress tolerance for improved fermentation performance. Quality considerations in brewing center on flavor stability through prevention of oxidative off-flavors such as cardboard and metallic tastes during storage, color preservation by minimizing melanoidin formation and enzymatic browning, and shelf-life extension through reduced oxidative deterioration via antioxidant management. Process innovations include reduced oxygen brewing techniques that minimize oxygen exposure throughout the entire brewing process, strategic antioxidant additions using ascorbic acid, sulfites, and natural antioxidants, and package protection systems employing oxygen-barrier materials and antioxidant packaging technologies.

Wine production requires a careful balance of oxidative and reductive conditions, with sulfite management serving dual roles as both antioxidant and antimicrobial agent. Controlled micro-oxygenation provides precise oxygen exposure during aging for optimal development, while temperature control reduces oxidative stress at lower fermentation temperatures. Quality enhancement in winemaking focuses on phenolic preservation to maintain the antioxidant capacity of grape polyphenols, controlled aroma development through managed oxidative reactions for complexity, and aging potential through oxidative stability for long-term storage. Distilled spirits production involves yeast stress management to enhance alcohol tolerance through oxidative stress control, controlled congener formation via managed oxidative reactions for flavor development, and consideration of oxidative stability during heating processes in distillation.

Dairy fermentation systems present unique challenges for oxidative stress management due to the sensitivity of starter cultures and the importance of maintaining product quality throughout shelf life. In yogurt and fermented milk production, starter culture protection involves antioxidant supplementation with ascorbic acid and α -tocopherol additions to milk, oxygen exclusion through modified atmosphere packaging and processing, and enhanced probiotic viability through improved stress tolerance for extended shelf-life. Process optimization includes temperature control to minimize heat-induced oxidative stress, pH management for optimal acidification rates to prevent stress, and packaging innovations using light and oxygen barrier materials. Cheese production requires managing oxidative processes during maturation, with controlled oxidative enzyme activity for lipid oxidation and flavor development, redox-mediated changes in microbial populations during succession, and protein oxidation effects on texture formation. Quality assurance measures include rancidity prevention through controlled lipid oxidation in high-fat cheeses, color stability by preventing oxidative discoloration, and balanced flavor development through managing both oxidative and reductive flavor compounds.

TABLE 1 | Oxidative stress features and control strategies across representative food fermentation systems.

Fermentation system	Key microorganisms	Major oxidative stress sources	Common biomarkers or indicators	Process or quality impact	Validated or potential control strategies
Beer and wine fermentation	<i>Saccharomyces cerevisiae</i> , non- <i>Saccharomyces</i> yeasts, lactic acid bacteria in some systems	Wort or must aeration, oxygen exposure during processing or aging, ethanol accumulation, nutrient limitation	DO, ORP, yeast viability, GSH/GSSG, antioxidant enzyme activity, lipid oxidation markers	Fermentation kinetics, yeast survival, aroma formation, oxidative off-flavors, color stability, shelf life	Controlled aeration, oxygen exclusion during storage, sulfite management in wine, yeast strain selection, antioxidant packaging
Dairy fermentation	Lactic acid bacteria, probiotic cultures, cheese-ripening microbiota	Oxygen exposure, light exposure, lipid oxidation, cold storage stress, redox changes during ripening	ORP, viable cell counts, MDA, protein oxidation markers, antioxidant enzyme activity	Starter activity, probiotic viability, acidification, texture development, rancidity, shelf life	Oxygen-barrier packaging, optimized starter selection, protective cultures, food-grade antioxidants, controlled ripening conditions
Fermented vegetables	Lactic acid bacteria, yeasts, acetic acid bacteria in some systems	Oxygen ingress, salt stress, plant phenolics, microbial H ₂ O ₂ production, redox-active plant metabolites	ORP, pH, microbial succession, H ₂ O ₂ , antioxidant capacity, color indicators	Acidification stability, microbial safety, color preservation, texture retention, off-flavor formation	Brine and salt control, anaerobic fermentation, temperature control, starter-guided fermentation, oxygen exclusion
Soy-based and mold-fermented foods	<i>Aspergillus oryzae</i> , <i>Rhizopus</i> spp., <i>Bacillus</i> spp., yeasts, lactic acid bacteria	Aerobic mold growth, lipid oxidation, long-term aging, Maillard-related redox reactions, oxygen gradients	ORP, volatile oxidation products, antioxidant capacity, enzyme activity, microbial community shifts	Proteolysis, amino acid metabolism, browning, aroma complexity, off-flavor formation, long-term aroma stability	Starter selection, controlled aeration, aging control, raw material quality control, matrix antioxidant management
Cereal and grain-based fermentations	Yeasts, lactic acid bacteria, <i>Bacillus</i> spp., filamentous fungi depending on product	Oxygen exposure, osmotic stress, organic acid accumulation, cereal phenolics, heating-derived pro-oxidants	ORP, pH, microbial viability, antioxidant capacity, ROS-related gene expression	Acidification, flavor development, texture, nutritional quality, fermentation reproducibility	Starter culture optimization, oxygen management, temperature control, substrate pretreatment, process standardization
Mixed and spontaneous fermentations	Lactic acid bacteria, yeasts, acetic acid bacteria, <i>Bacillus</i> spp., filamentous fungi	Interspecies ROS production, oxygen gradients, microbial succession, H ₂ O ₂ production, variable raw materials	ORP, DO, H ₂ O ₂ , catalase or peroxidase activity, community profiling, microbial viability	Community stability, acidification, flavor development, safety, batch-to-batch consistency	Starter-guided fermentation, oxygen control, microbial community monitoring, process standardization, matrix-specific redox management

Fermented vegetables represent traditional fermentation systems where oxidative stress management occurs naturally through microbial community interactions. In sauerkraut and kimchi production, spontaneous fermentation management involves controlling ROS-mediated changes in community composition during microbial succession, maintaining antioxidant vitamin content for nutrient preservation, and ensuring safety through pathogen inhibition via controlled ROS production. Process optimization requires balancing salt concentration to maintain preservation while supporting microbial activity, establishing anaerobic conditions to minimize oxygen exposure during fermentation, and temperature control for optimal conditions supporting beneficial microorganisms. Pickled products require careful fermentation environment control through brine composition management including salt concentration and antioxidant additions, pH management for appropriate acidification rates and final pH targets, and oxygen exclusion to maintain anaerobic fermentation conditions. Quality maintenance focuses on color preservation by preventing oxidative browning and discoloration, texture retention through minimizing oxidative softening of plant tissues, and nutritional quality preservation of vitamins and antioxidant compounds.

Beyond vegetable fermentations, soy-based fermentations provide additional examples in which oxidative stress influences both microbial activity and product quality. In products such as tempeh, miso, soy sauce, and natto, oxidative stress may influence the activity of *Rhizopus* spp., *Aspergillus oryzae*, *Bacillus* spp., yeasts, and lactic acid bacteria, thereby affecting proteolysis, lipid oxidation, amino acid metabolism, volatile generation, browning reactions, and long-term aroma stability. These systems highlight that oxidative stress management in food fermentation should be considered at both microbial and matrix levels.

Plant-based and cereal-based fermentations provide further examples in which oxidative stress is linked to both microbial physiology and matrix-derived redox chemistry. For example, oat bran fermentation by *Lactobacillus fermentum* NB02 with feruloyl esterase production has been reported to increase phenolic compound content and antioxidant properties, illustrating how microbial activity can reshape the redox-active matrix of cereal substrates (Y. Li et al. 2024). Oat phenolic compounds have also been associated with antioxidant activity and prebiotic effects, suggesting that plant-derived phenolics may influence both oxidative balance and microbial ecology. Therefore, oxidative stress management in plant-based fermentations should consider not only microbial stress tolerance, but also phenolic release, matrix antioxidant capacity, prebiotic potential, and flavor development.

Traditional fermented foods from various cultures also demonstrate empirical strategies for oxidative stress management. Asian fermented foods include soy-based products, sake, rice wine, and other rice-based fermentations, in which redox balance may affect mold growth, yeast fermentation, aging stability, enzyme activity, and aroma development. African and Latin American fermented foods, including cassava- and maize-based products such as gari, fufu, pozol, atole, and boza, further show that oxidative stress management can be linked to product

safety, nutritional preservation, flavor development, and process reproducibility.

In mixed-culture spontaneous fermentations, microbial succession is strongly influenced by oxygen availability, redox-active substrates, and interspecies stress interactions. Early oxygen consumption by yeasts, facultative bacteria, or aerobic members can shift the system toward a more reduced environment, favoring the growth of lactic acid bacteria and other oxygen-sensitive organisms. At the same time, ROS or H₂O₂ produced by certain bacteria may inhibit susceptible competitors and reshape community assembly. These redox-mediated interactions contribute to community-level resilience, but they can also increase batch-to-batch variability when raw materials, oxygen exposure, salt concentration, or temperature are not well controlled. Therefore, oxidative stress management in spontaneous fermentations should consider microbial succession, interspecies redox balance, flavor development, safety, and fermentation reproducibility together. Taken together, these traditional and spontaneous fermentation systems demonstrate that oxidative stress management has been empirically shaped across cultures and can provide useful insights for modern fermentation technology development.

8 | Future Perspectives and Challenges

The field of oxidative stress management in fermentation is rapidly evolving, driven by advances in monitoring technologies, synthetic biology, and systems-level understanding that could contribute to fermentation optimization while presenting new challenges requiring innovative solutions. Emerging technologies may improve the ability to monitor and control oxidative stress in real-time through advanced sensor systems and data integration platforms. Electrochemical biosensors are being developed to provide direct, real-time ROS detection in fermentation broth, while optical sensors offer fluorescence-based monitoring with improved sensitivity for detecting specific oxidative stress markers. Spectroscopic methods such as NIR and Raman spectroscopy enable noninvasive redox state assessment without disrupting fermentation processes, and microfluidic devices are emerging as powerful lab-on-chip systems for rapid, multi-parameter analysis that can provide comprehensive oxidative stress profiles from minimal sample volumes. Data integration capabilities are advancing through machine learning algorithms that enable predictive modeling for oxidative stress management, similar to successful applications in other bioprocess optimization areas. Process analytical technology (PAT) systems provide real-time control and optimization capabilities, while digital twins create virtual fermentation models for process optimization that can predict oxidative stress development before it becomes problematic. Internet of Things (IoT) networks connect multiple sensors for comprehensive monitoring across entire production facilities, enabling coordinated responses to oxidative stress events. Wireless and remote monitoring capabilities include sensor miniaturization for implantable continuous monitoring devices, cloud-based analytics for remote data processing and analysis, and mobile applications that enable real-time process monitoring and control.

For practical implementation, oxidative stress management strategies should be distinguished according to their current readiness. Established or near-term practices include oxygen control, ORP/DO monitoring, starter culture selection, process optimization, food-grade antioxidant management, and packaging-based oxygen control. In contrast, synthetic biology, nanotechnology, digital twins, and IoT-based control remain emerging approaches in food fermentation. Their broader application requires validation in real food matrices, cost-benefit evaluation, scalability assessment, regulatory approval, and clear evidence linking redox control with microbial performance and product quality.

Synthetic biology applications are opening new frontiers in oxidative stress management through engineered biological systems that can both detect and respond to oxidative conditions with greater flexibility. Biosensor development includes genetically encoded indicators using fluorescent proteins responsive to specific ROS species, similar to calcium indicators that have revolutionized cell biology research. Whole-cell biosensors function as living environmental monitoring systems that can provide continuous feedback on fermentation conditions, while synthetic circuits provide programmable stress responses and adaptive behaviors that can be precisely tuned for specific applications. Metabolic engineering approaches focus on pathway optimization for enhanced antioxidant production and ROS scavenging, building on successful examples like enhanced glutathione production in *S. cerevisiae* and improved catalase expression in *Aspergillus niger*. Stress tolerance engineering aims to improve robustness under oxidative conditions through approaches such as overexpression of SOD and catalase, similar to successful strategies used in developing ethanol-tolerant yeast strains. Advanced synthetic biology tools include CRISPR-based systems for precise genetic modifications that enhance stress tolerance, protein design for engineered enzymes with enhanced antioxidant activity, and regulatory circuit design for tunable stress response systems.

Nanotechnology applications are emerging as promising tools for oxidative stress management, offering unprecedented precision in antioxidant delivery and monitoring capabilities. Nanoparticle-based antioxidants provide targeted delivery systems for specific antioxidant delivery to cellular compartments, overcoming limitations of traditional supplementation approaches. Sustained release formulations offer long-term protection throughout extended fermentation processes, while enhanced bioavailability improves antioxidant uptake and utilization efficiency. Nanosensors enable single-cell analysis for individual cell oxidative stress monitoring, providing insights into population heterogeneity that bulk measurements cannot capture, and support high-throughput screening for rapid assessment of multiple conditions. Similarly, nanoparticle-based antioxidants, nanosensors, and nanomaterial-based delivery systems should be considered exploratory in food fermentation until their migration behavior, toxicity, residue risk, food-contact safety, environmental impact, and regulatory compliance are systematically evaluated.

Food safety and regulatory considerations are essential for translating oxidative stress management strategies into food

fermentation applications. Antioxidant supplementation should prioritize food-grade compounds with established safety profiles, and its dosage, sensory impact, labeling requirements, and regulatory status should be evaluated for each product category. Engineered or genetically modified strains may enhance oxidative stress tolerance, but their application in foods requires careful assessment of genetic stability, containment, horizontal gene transfer risk, regulatory approval, and consumer acceptance. Similarly, nanosensors and nanomaterial-based antioxidant delivery systems remain exploratory in food fermentation until their migration behavior, toxicity, residue risk, food-contact safety, environmental impact, and regulatory compliance are systematically evaluated. Therefore, emerging redox management technologies should be considered applicable to food fermentation only after validated risk assessment, process-specific demonstration, and compliance with relevant food safety regulations.

Future research should prioritize four process-oriented directions that can move oxidative stress research from descriptive monitoring toward decision-support systems for food fermentation. First, real-time and noninvasive redox monitoring methods should be validated in complex food fermentation matrices rather than only in model systems. Such methods should account for matrix interference, electrode stability, sampling disturbance, and the need to link redox signals with measurable fermentation outcomes. Second, strain- and community-level heterogeneity in oxidative stress responses should be resolved using single-cell, spatial, and multi-omics approaches, particularly in mixed and spontaneous fermentations where different microorganisms experience distinct oxygen and redox microenvironments. Third, redox biomarkers should be integrated with PAT to support adaptive process control; however, process-specific thresholds must be established before industrial implementation. Fourth, predictive models should be developed to link redox status, microbial physiology, community succession, fermentation performance, and product quality endpoints. These priorities would help shift oxidative stress research from descriptive monitoring toward evidence-based management in real food fermentation systems. Industrial implementation presents significant challenges requiring innovative solutions, particularly in translating laboratory findings to commercial production scales. Scale-up challenges involve addressing spatial oxidative stress gradients in large bioreactors where mixing limitations can create zones of varying oxygen concentration, developing consistent stress control across different scales, and creating cost-effective mitigation strategies that do not compromise economic viability. Technology transfer requires pilot-scale validation to bridge laboratory and industrial scales, process integration to incorporate oxidative stress management into existing systems without major infrastructure changes, and regulatory considerations to meet safety and quality requirements. Sustainability considerations focus on energy efficiency to reduce energy consumption in oxidative stress management, waste minimization through improved process yields, and green chemistry approaches using sustainable antioxidant sources. Personalized fermentation approaches involve strain-specific optimization through individual strain characterization, customized protection strategies with tailored antioxidant systems, and adaptive process control for real-time adjustment based on strain behavior.

Although these technologies may provide useful tools for redox management, their relevance to food fermentation depends on validation in real food matrices, compatibility with food-grade processing, regulatory acceptance, and demonstrated links with microbial performance and product quality.

9 | Conclusion

Oxidative stress represents a fundamental aspect of microbial fermentation that significantly influences process efficiency, product quality, and industrial sustainability. This comprehensive review has highlighted the dual nature of oxidative stress, demonstrating how controlled ROS levels serve essential signaling functions while excessive accumulation leads to cellular damage and fermentation failure. The mechanistic understanding reveals that sources of oxidative stress in fermentation systems are diverse, encompassing both endogenous metabolic processes such as respiratory chain electron leakage, enzymatic ROS generation, metabolic flux imbalances, and product accumulation feedback, as well as exogenous environmental factors including physico-chemical stressors, substrate-derived compounds, and microbial interactions. Understanding these sources enables targeted intervention strategies that can prevent harmful ROS accumulation while preserving beneficial signaling functions. The dichotomous impact of ROS underscores the importance of precise control rather than complete elimination, where physiological levels promote beneficial adaptive responses, metabolic reprogramming, and cross-protection, while toxicological levels cause detrimental macromolecular damage and fermentation inhibition.

Detection and monitoring capabilities have advanced significantly through the development of sophisticated analytical techniques including fluorescent probes, biomarker analysis, and real-time monitoring systems that provide unprecedented insight into fermentation redox status. Multi-omics approaches combining transcriptomics, proteomics, and metabolomics enable comprehensive understanding of stress responses and facilitate proactive process control and optimization. These monitoring capabilities are essential for implementing effective mitigation strategies that integrate process engineering approaches such as oxygen management and feeding optimization, strain development including antioxidant system enhancement and stress response optimization, and medium formulation strategies involving antioxidant supplementation and pro-oxidant minimization.

Industrial applications across diverse fermentation processes demonstrate the practical value and broad applicability of oxidative stress control in food production. From alcoholic beverage production where controlled oxidation contributes to flavor development, to dairy fermentations where oxidative stress management preserves starter culture viability, to traditional fermented foods where natural antioxidant systems have evolved over centuries, successful implementation shows that oxidative stress management can improve product quality while maintaining process efficiency. These applications provide valuable case studies for developing new approaches and scaling existing technologies to industrial production levels.

Future research should focus on establishing evidence-based links between redox indicators, microbial physiology, community succession, fermentation performance, and product quality endpoints. Real-time and multiparameter monitoring systems, predictive modeling, strain improvement, and adaptive process control may provide useful tools for oxidative stress management, but their practical value depends on validation in real food matrices, process-specific calibration, and demonstration at pilot or industrial scale. In particular, ORP, DO, ROS biomarkers, and antioxidant capacity should be interpreted together with measurable outcomes such as microbial viability, fermentation kinetics, volatile formation, color stability, texture retention, safety, and shelf life.

Overall, oxidative stress management provides a promising conceptual and practical framework for improving food fermentation, but its implementation should remain process-specific and evidence-based. Established or near-term practices, such as oxygen control, ORP/DO monitoring, starter culture selection, process optimization, food-grade antioxidant management, and packaging-based oxygen control, are already relevant to many fermentation systems. In contrast, emerging approaches, including engineered stress-response circuits, nanosensors, nanomaterial-based antioxidant delivery, digital twins, and IoT-based control, require further validation in real food matrices, food safety assessment, regulatory evaluation, cost-effectiveness analysis, and industrial-scale demonstration. Future progress will depend on separating established practices from exploratory technologies and on linking redox management with measurable improvements in microbial performance, product quality, and fermentation reproducibility.

Author Contributions

Wanrong Hu: conceptualization, writing – original draft, methodology. **Manxi Wu:** visualization, writing – original draft, software. **Hui Xu:** investigation, data curation. **Jiebiao Chen:** investigation, software. **Wen Cai:** resources, validation. **Han Yang:** data curation. **Hujing Cao:** formal analysis. **Xinlin Huang:** validation. **Jinping Cao:** writing – review and editing. **Chongde Sun:** supervision, formal analysis. **Yue Wang:** project administration, conceptualization, supervision. **Dongliang Li:** funding acquisition, resources.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

No new data were generated or analyzed in this study. All information discussed in this review is available from the cited publications.

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